

Lecture 5

Analogue to Digital Conversion

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Topics Covered (Chapter 7)

- Sampling Theorem
- Quantization Noise
- Uniform Quantization
- ADC specifications
- DAC

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7-3: Data Conversion

- Translating an analog signal into a digital signal is called **analog-to-digital (A/D) conversion**, **digitizing a signal**, or **encoding**.
 - The device used to perform this translation is known as an **analog-to-digital converter** or **ADC**.
- Translating a digital signal into an analog signal is called **digital-to-analog (D/A) conversion**.
 - The circuit used to perform this is called a **digital-to-analog (D/A) converter** or **DAC** or a **decoder**.

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7-3: Data Conversion

- An analog signal is a smooth or continuous voltage or current variation.
- Through A/D conversion these continuously variable signals are changed into a series of binary numbers.
- A/D conversion is a process of **sampling** (or measuring the analog signal at regular time intervals), **quantisation** and **coding**.

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7-3: Data Conversion

To retain the high-frequency information in the analog signal, a sufficient number of samples must be taken to adequately represent the waveform.

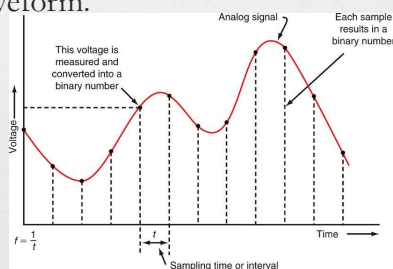


Figure 7-7: Sampling an analog signal

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Sampling Theorem

- How densely should we sample an analogue signal so that we can reproduce its form accurately?
- Nyquist Sampling Theorem: **A signal the spectrum of which is band-limited to f_m Hz, can be reconstructed exactly from its samples, if they are taken uniformly at a rate of $f_s \geq 2f_m$ Hz.**
- When $f_s < 2f_m$ Hz, the signal is undersampled.
- When $f_s > 2f_m$ Hz, the signal is oversampled.

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Sampling Theorem

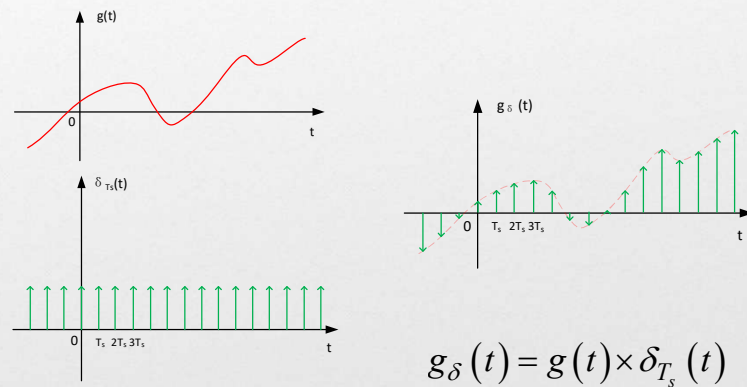
- Relationship between the **Nyquist rate** (f_s) and the maximum analogue frequency (f_m)
- The sampled signal, $g_\delta(t)$ is the product of the analogue signal, $g(t)$, and the switching signal, $\delta_{T_s}(t)$,

$$g_\delta(t) = g(t) \times \delta_{T_s}(t)$$
- The result is given as the sum of the original signal and the product of the original signal and an infinite array of sinusoids of frequencies $f_s, 2f_s, 3f_s, 4f_s, \dots$

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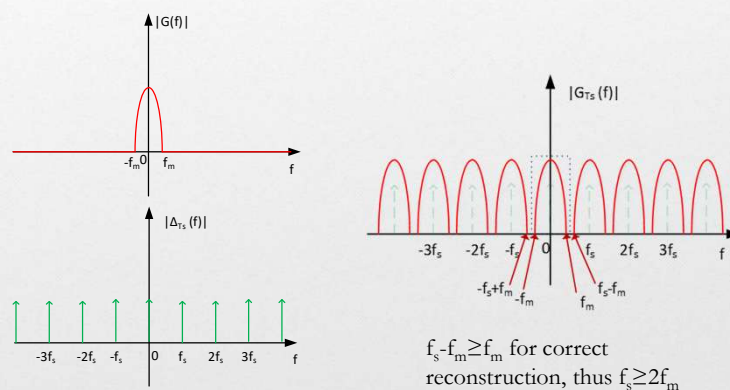
Sampling: Time Domain



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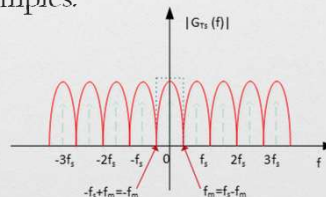
Sampling: Frequency Domain



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Critical Sampling

- When $f_s = 2f_m$, the sampling frequency is exactly equal to the Nyquist Rate, it is called critical sampling.
- At critical sampling, the sampling interval $T_s = 1/(2f_m)$ is called the Nyquist interval, and the sampled signal is called the Nyquist samples.



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Undersampling

- When $f_s < 2f_m$, it is called undersampling.
 - The sidebands in the spectrum overlap, resulting in **aliasing** distortion.
 - Every frequency component f_h in the analogue signal higher than half the sampling frequency appears in the sampled signal at a false or alias frequency. $f_a = |f_s - f_h|$.
 - The reconstruction filter will pass two signals: the analogue signal and the alias signal.

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Aliasing

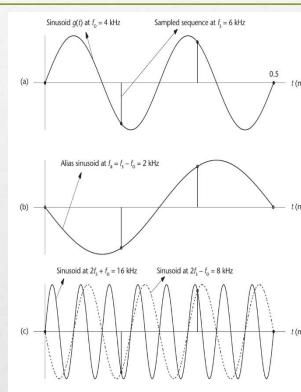
- Aliasing causes a new signal to be created near the original signal.
- This signal has a frequency of $f_a = f_s - f_m$
- When the sampled signal is converted back to analogue signal by a D/A converter, the output will be the alias, not the original signal.

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Aliasing—An Example

- Consider sampling a 4kHz sine-wave at 6000 sample/sec. (below Nyquist rate)
- Produces an alias frequency of $f_a = f_s - f_m = 2$ kHz
- Also produces frequencies of: $f_m, f_s \pm f_m, \dots$
- Both f_a and f_m will pass through the reconstruction filter.



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Solution to Aliasing

- To eliminate this problem, a low-pass filter called an **anti-aliasing filter** is usually placed between the modulating signal source and A/D converter input.
- The anti-aliasing filter ensures that no signal with a frequency greater than one-half the sampling frequency is passed.
- This filter must have extremely good selectivity.

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Solution to Aliasing

- Anti-aliasing low-pass filter removes the non-essential high-frequency component in message signal.
 - Telephony, $f_{\text{cut-off}} < 4$ kHz for a sampling frequency of 8 kHz
 - Music, $f_{\text{cut-off}} < 20$ kHz for a sampling frequency of 44.1 kHz.
- High order active filter is used because it has steep roll-off
- Usually, this transmit conditioning LPF is identical to the receive reconstruction filter.

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Oversampling

- When $f_s > 2f_m$, it is called oversampling.
 - When $G(f)$ contains an impulse at the highest frequency $\pm f_m$, the sampling rate f_s must be greater than $2f_m$ Hz.
 - A well-known example is an sinusoid $g(t) = \sin[2\pi f_m(t-t_0)]$, which is band-limited to f_m Hz, but all its samples are zero when uniformly taken at a rate $f_s = 2f_m$ (starting from $t=t_0$), and $g(t)$ cannot be recovered from its Nyquist samples.
 - Thus, for sinusoids, the condition $f_s > 2f_m$ must be satisfied.

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7-3: Data Conversion

A/D Converters

- A/D conversion begins with sampling, which is carried out by a **sample-and-hold (S/H) circuit**.
- The S/H circuit takes a precise measurement of the analog voltage at specified intervals.
- The A/D converter then converts this instantaneous value of voltage and translates it to a binary number.

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7-3: Data Conversion

A/D Converters: S/H Circuits

- A sample-and-hold (S/H) circuit, also called a **track/store circuit**, accepts the analog input signal and passes it through, unchanged, during its sampling mode.
- In the hold mode, the amplifier remembers or memorizes a particular voltage level at the instant of sampling.
- The output of the S/H amplifier is a fixed DC level whose amplitude is the value at the sampling time.

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7-3: Data Conversion

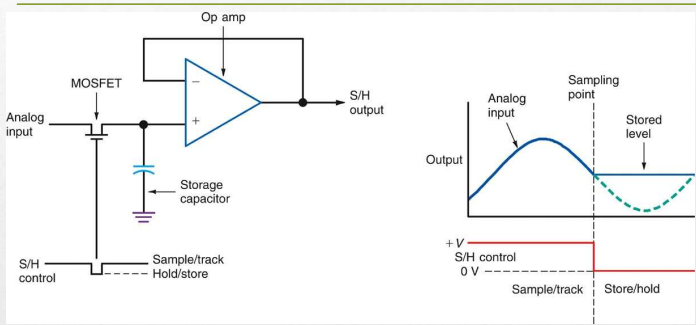


Figure 7-18: An S/H amplifier

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7-3: Data Conversion

A/D Converters: S/H Circuits

- The primary benefit of an S/H amplifier is that it stores the analog voltage during the sampling interval.
- In some high-frequency signals, the analog voltage may change during the sampling interval.
 - This is undesirable because it introduces **aperture error**.
- The S/H amplifier stores the voltage on the capacitor. With the voltage constant during the sampling interval, quantizing is accurate.

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Quantisation

- The analog signal represents an infinite number of actual voltage values. The A/D converter can represent only a finite number of voltage values over a specific range.
- The sampled signal, although discrete in time, is still continuous in value, and thus is not a digital signal.
- The quantisation process converts the continuous sample amplitudes to a finite set of discrete values.

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Quantisation

- The samples are converted to a binary number whose value is close to the actual sample value.
- A/D converter divides a voltage range into discrete increments, each represented by a binary number.
- The analog voltage measured during the sampling is assigned to the voltage increment closest to it.
- Errors associated with this process are known as **quantizing errors**.

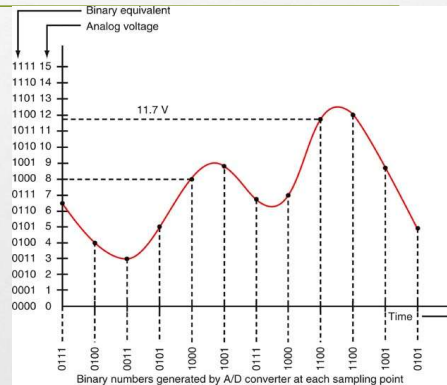
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Uniform Quantisation

- Vertical axis: known as the quantisation scale. Linear scaling uniform step size. Number of steps is given by $N=2^K$ steps, where K is number of bits used to represent one sample.
- Sample is compared to the scale and rounded to the nearest integer value.
- The quantised value may be then transmitted as a binary number.

Figure 7-8: The A/D converter divides the input voltage range into discrete voltage increments.



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Uniform Quantisation

- All intervals or steps are of the same size.
 - Quantiser step size = Δ
- Each sample is rounded to the nearest integer value Q .
- The maximum quantisation error is $\Delta/2$ (half the step size).
- To reduce the quantisation noise, reduce the step size, however more bits are required.
 - K bits give N steps ($N=2^K$)
 - $K=7$ bits gives $N=128$ steps; $K=8$ bits gives $N=256$ steps.
 - For a given dynamic range, the step size decreases.

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Quantisation Noise

- *Mean square quantization error* (MSQE) is a figure of merit for the process of quantisation. MSQE is found by integrating the probability that square quantisation error falls within the step Δ .
 - This gives $\text{MSQE} = \Delta^2/12$
 - MSQE only depends on the step size.
- If the quantisation scale ranges from $-C$ to $+C$ then
 - $\Delta = 2C/N = 2C/2^K$
- Quantisation noise can be reduced by increasing the number of bits per sample.
- This increases the bit rate and hence the bandwidth.

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Quantisation Noise

- Equation for the sawtooth
 $e(t) = st$ for $\Delta/2s < t < \Delta/2s$

- Mean square value is

$$\frac{1}{b-a} \int_a^b x^2 dx$$

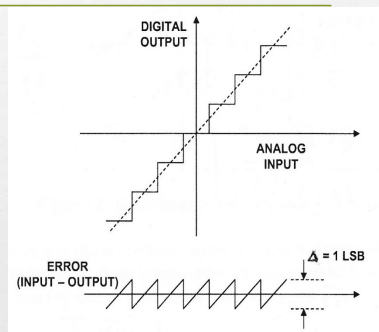
therefore

$$\overline{e^2(t)} = \frac{1}{\frac{\Delta}{2s} + \frac{\Delta}{2s}} \int_{-\frac{\Delta}{2s}}^{\frac{\Delta}{2s}} (st)^2 dt$$

Which gives

$$\sqrt{\overline{e^2(t)}} = \frac{\Delta}{\sqrt{12}}$$

the RMS quantisation noise



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Dynamic Range

- Largest signal amplitude
 - An analogue signal of amplitude $V_{\text{peak}} = C$ will fully load the quantiser, i.e. the signal will cover the full range from $-C$ to $+C$.
 - Therefore $V_{\text{max}} = 2C$
- Smallest signal amplitude
 - Analogue signal of amplitude $< \Delta$ will not change to next step
 - Therefore $V_{\text{min}} = \Delta$
- Dynamic Range (DR) = $V_{\text{max}}/V_{\text{min}} = 2C/\Delta = 2^K$
 - Changing this to decibels using $20 \log_{10}(V_{\text{max}}/V_{\text{min}})$ gives dynamic range = 6.02K dB (for an 8-bit ADC, DR = 48.2 dB)
 - Typically the dynamic range for speech is 60 dB.

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Signal to Quantisation Noise Ratio

- An important parameter for assessing the performance of quantiser.
- SQNR = signal power / MSQE
- Can be reduced to:

$$SQNR = \frac{V_p^2/R^2}{\Delta^2/12}$$

$$\begin{aligned} SQNR &= 10\log(3) + 10\log(2^{2K}) + 10\log(R^{-2}) + 10\log(V_p^2/C^2) \\ &= 4.77 + 6.02K - 20\log(R) + 20\log(V_p/C) \text{ dB} \end{aligned}$$

where R = peak to RMS ratio of the signal (for a sine wave $R = 1.414$)

- If $V_p = C$, then $SQNR = 4.77 + 6.02K - 20\log(R)$ dB.

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SQNR

- For a sinusoidal test tone that fully loads the quantiser
 - $V_p = C$; and $R = \text{peak to RMS ratio} = 1.414$
- $\text{SQNR} = 4.77 + 6.02K - 20\log(1.414) \text{ dB}$.
 $= 4.77 + 6.02K - 3.01$
 $= 1.76 + 6.02K \text{ dB}$
- For an 8 bit system, $\text{SQNR} = 50\text{dB}$, and gets worse if the quantiser is under-loaded. While this figure is acceptable for telephony, it is not acceptable for music.

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SQNR

- In telephony, it is desirable to have a constant SQNR over a wide range of input signal values so that service quality is maintained at same level for both quiet and loud talkers.
- Number of bits (K) used must be kept to an acceptable low value (for telephony $K = 8$ bits) to ensure that the transmission bandwidth is not too large.
- Uniform quantisation is replaced by non-uniform quantisation: step size varies across the quantisation scale.

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ADC Specifications

- The key ADC specifications are
 - Resolution
 - Dynamic range
 - Signal-to-noise ratio
 - Effective number of bits
 - Spurious free dynamic range.

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ADC Specifications

- **Resolution** is related to the number of bits and indicates the smallest input voltage recognized by the converter. It is the reference voltage V_{REF} divided by 2^K , where K is the number of output bits.
- **Dynamic range** is a measure of the input voltage range that can be converted.
- **Signal-to-noise ratio (SNR)** is the ratio of actual input signal voltage to the total noise in the system.

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ADC Specifications

- **Spurious free dynamic range (SFDR)** is the ratio of RMS signal voltage to highest “spur” voltage expressed in decibels.
 - A **spur** is any spurious or unwanted signal that may result from intermodulation distortion.
- Noise, harmonics, or spurious signals all add together and reduce the resolution of an ADC. This effect is expressed by a measure called **effective number of bits (ENOB)**

$$\text{ENOB} = \frac{\text{SINAD} - 1.76}{6.02}$$

SINAD: signal-to-noise-and-distortion ratio in dB

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ADC Specifications

- Common ways to translate analog voltage to binary number:
 - **Successive-Approximations Converters:** iteratively compare input to a series of reference voltages and refine output bit-by-bit
 - **Flash Converters:** use a bank of comparators to compare input to multiple reference levels and produce output in a single step
 - **Pipelined Converters:** divide conversion process into multiple stages, each processing a portion of output sequentially
- Binary numbers representing the samples can be stored in random access memory (RAM), disk, or magnetic tape.
 - They can then be processed and used as data by computers that perform mathematical and logical manipulations.

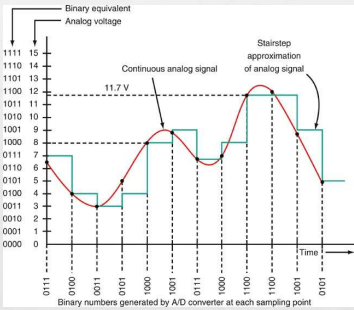
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DAC

- The D/A converter receives the binary numbers sequentially and produces a proportional analog voltage at the output.

Figure 7-9: A D/A converter produces a stepped approximation of the original signal.



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